BUSINESS/IT
REQUIREMENTS DOCUMENT

Smart Water Management Prediction System
SWMPS-2025

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IT Project Manager:	IT Lead
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DOCUMENT CONTROL

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Document Approval

Approver	Title	Business Area	Approval Date
Mr. Amit Shah	Maintenance Lead	Field Maintenance	

Document Distribution

Name	Title	Business Area
Mr. Anish Sharma	Data Scientist	Analytics Team

Revision History

Date	Author	Version	Summary Of Changes	Status
30/12/2025	Aman Sharma	0.1	Initial Draft	Draft
31/12/2025	Aman Sharma	1.0	Final with FR/NFR/Flowcharts	Released

Related Documents

Document Name	Version	Author	Date
Smart Water Capstone Project	1.0	Training Team	31/12/2025
BRD Template	1.0	BA Standards	2025

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1 Introduction

The Smart water Management Prediction System Automates the monitoring of more than 100 water pumps using sensors and ML to predict failures 48hrs early, reducing breakdown by 75% and saving Saving Rs.5 Lacs a year in water pump repairing.

2 Problem/Impact/Successful Outcome

The Problem	The Impact	The Successful Outcome
Manual Monitoring more than 100 water pumps leads to missing early sings of failures.	Monitoring manually 100+ pumps is time consuming and because of missing early signs of water pump failures, gives costly repairs charges of $50K+.	An Automated Monitoring will give 48hrs early real time alerts, which will take less manpower, help to cutting down the costly expenses up to 75%.

3 Objectives

ID	Business Objective	Business Owner	Business Importance
001	Reduce Pump Failures by 75%	Operations Manager	High
002	Automate the pump monitoring 24/7	Maintenance Engineers	High
003	Provide Real time Dashboard	Operations Manager	Medium
004	Reduce repair costs 50%	Finance	High
005	99% pump uptime	Operations	High

4 Purpose Of Document

The Business Requirements Specification details the business requirements as elicited by the business analyst from the key stakeholders. The document presents the requirements in a structured way that facilitates review and sign off by the designated approvers.

Building on the high level scope of the project as defined in the Project Definition Document, the business requirements clearly state in business language what any chosen solution must do.
This document captures the business requirements in a structured way, providing the basis for ensuring that the solution delivered meets the requirements.

It should:-

- Facilitate a shared understanding for all stakeholders of the business requirements
- Be the key input for the preparation of a functional requirements specification
- Facilitate the identification of possible solutions

5 Scope

In Scope	Out Of Scope
Sensor Data collection, ML Prediction, Alerts, Dashboards, User Roles.	Hardware Installation, Pump Manufacturing, Billing system.

6 Definitions, Acronyms and Abbreviations

Abbreviation/Acronym	Description
ML	Machine Learning
SMS	Short Message Service
UI	User Interface
NFR	Non-Functional Requirement

7 Risks

Ref	Risk	Detailed BRS Reference
R01	Sensor Hardware Failure	FR-01
R02	ML Model accuracy drops	FR-02
R03	Data privacy breach	FR-03

8 Assumptions

Ref	Assumption	Detailed BRS Reference
A01	Sensors Installed on all pumps	
A02	Internet Connectivity at pump sites	

9 Issues

Ref	Issue	Detailed BRS Reference
I01	Sensor calibration standards	FR01

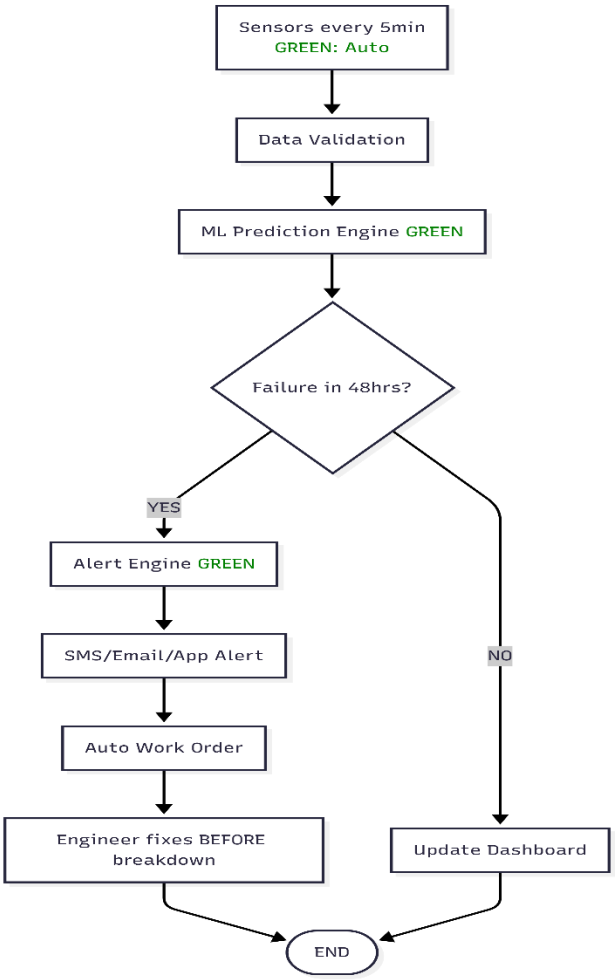
10 Dependencies

Ref	Dependency	Detailed BRS Reference
D01	Sensor hardware delivery	FR-01
D02	Weather API contract	FR-08

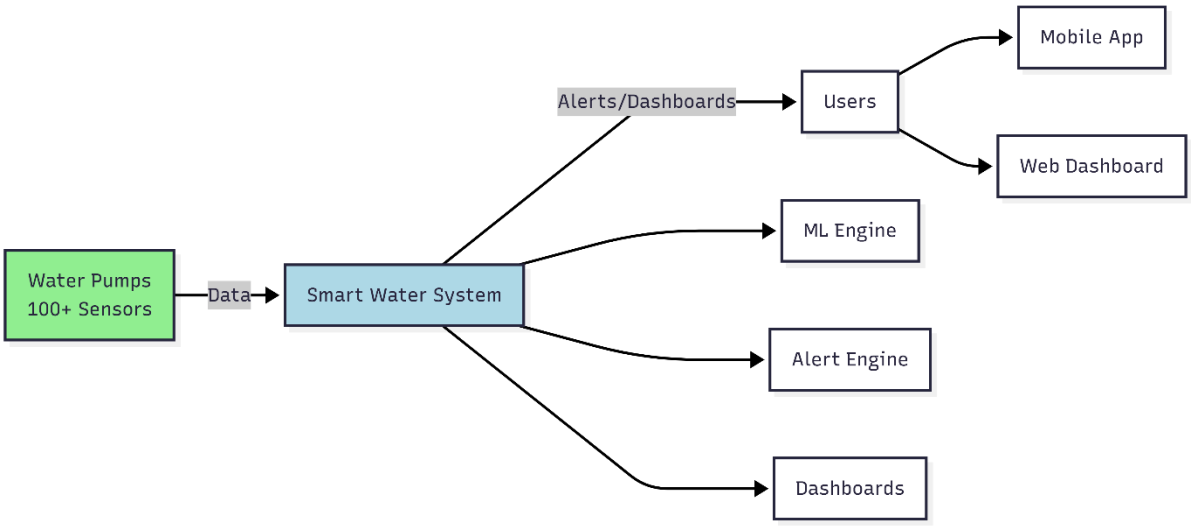
11 As Is Process

- Engineers drive to pump sites daily
- Manually reads gauges, notes issues
- Reports to manager via phone/email
- Manager decides repairs priority

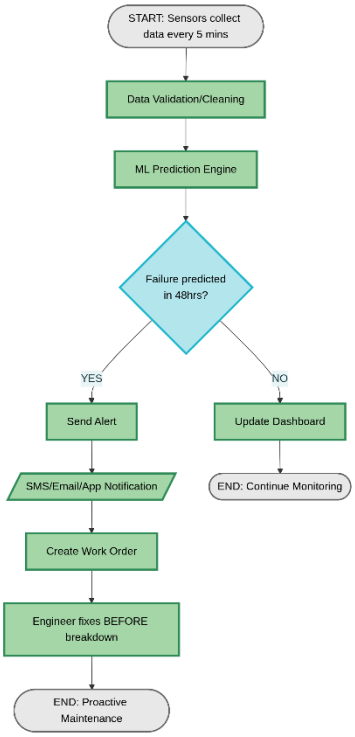
Note: Slow (48hrs per site visits), may misses early warnings, reactive fixes.



12 Context Diagram



13 Process Overview Diagram



14 High Level To Be Business Requirements

- Collaborative documentation using Confluence for real – time stakeholder updates and version control.
- Real-time sensor monitoring system (flow, pressure, vibration)
- ML model to predicts failures 48hrs ahead.
- Automated alerts vis SMS/email/app.
- Role-bases dashboards and reports.

User Story:

User Story	Acceptance Criteria	Priority
As Operations Manager, I want real-time dashboards so I can monitor all 100+ pumps status.	Dashboard loads less than 3 seconds, shows failure predictions.	Must
As Maintenance Engineer, I want SMS alerts so I can fix pumps before breakdown.	Alerts within 2min, includes pump location.	Must
As Data Scientist, I want clean sensor data so I can train accurate ML models	Data validated, outliers removed	Should

15 Detailed Business/IT Requirements

15.1 Functional Requirements

ID	Title	Requirements Description	Type (*)	MoSCoW Priority	Originator	Status (**)	Delivered By	Test ID
FR-01	Real-time sensor monitoring	System should collect water flow, pressure, temperature from 100+ pumps in every 5minutes.	Application	Must/High	Operations Manager	Proposed	NA	NA
FR-02	Failure Prediction Alert	ML Model should predict the pump failure 48hrs ahead bases on pump pressure through sensors.	Application	Must/High	Maintenance Engineer	Proposed	NA	NA
FR-03	SMS/Email Alerts	Send Critical alerts to engineers within 2min of prediction	Integration	Must/High	Maintenance Engineer	Proposed	NA	NA
FR-04	Operations Dashboard	Manager views all pumps status, pressure, failure trends on single screen.	Application	Must/High	Operations Manager	Proposed	NA	NA
FR-05	Historical Reports	Generate monthly reports: failure rates, prediction accuracy, cost savings.	Information	Should/Medium	Operations Manager	Proposed	NA	NA
FR-06	User Role	Define role: Admin, Manager, Engineer, Viewer with Specific Dashboard Access.	Technical	Must/High	IT Admin	Proposed	NA	NA
FR-07	Data Validation	Validate sensor data before ML processing: remove outliers, fil gaps	Technical	Must/High	Data Scientists	Proposed	NA	NA
FR-08	Weather Integration	Pull local weather data to adjust predictions: rain increased demand	Technical	Must/Medium	Data Scientists	Proposed	NA	NA
FR-09	Mobile App Access	Engineers can view alerts and dashboards on Mobile app	Application	Should/Medium	Maintenance Engineers	Proposed	NA	NA
FR-10	Audit Logs	Track All alerts, predictions user actions for compliance	Audit	Must/High	Compliance Officer	Proposed	NA	NA
FR-11	Confluence Integration	Requirements stored in Confluence with version control, comments, @mentions	Technical	Must/High	IT Admin	Proposed	NA	NA

* Type Key: -

- Business
- Application
- Information
- Integration
- Technical

** Status Key: -

- Proposed
- Accepted

MoSCoW Rating: -

- **Must Have**: Describes a requirement that must be satisfied in the final solution for the solution to be considered a success
- **Should Have**: Represents a high-priority item that should be included in the solution if it is possible. This is often a critical requirement but one which can be satisfied in other ways if strictly necessary.
- **Could Have**: Describes a requirement which is considered desirable but not necessary. This will be included if time and resources permit.
- **Would Have**: Represents a requirement that stakeholders have agreed will not be implemented in a given release, but may be considered

15.2 Process Diagram

Details Process Flow Documented in:
Section 11: As is Manual Process
Section 13: To -Be Automated Process
Detailed flows in sections 11-13. All requirements tracked in Confluence with version history and stakeholder comments.
No Additional diagram required – flows link directory to FR-01 through FR-10 above.

15.3 Non-Functional Requirements

ID	Title	Requirements Description	Type (*)	MoSCoW Priority	Originator	Status (**)	Delivered By	Test ID
NFR-01	Performance	Dashboard loads in less than 3 seconds	Performance	High	Operations Manager	Proposed	NA	NA
NFR-02	Availability	99.9% uptime, Around 45min of downtime pm max.	Availability	High	Compliance Officer	Proposed	NA	NA
NFR-03	Security	Role-Based access, encrypted data transmission	Security	High	Maintenance Engineers	Proposed	NA	NA
NFR-04	Scalability	Supports 300+ pumps without performance drop.	Capacity	Medium	Maintenance Engineers	Proposed	NA	NA
NFR-05	Recovery	Full system recovery in <30 mins	Recovery	Medium	Data Scientist	Proposed	NA	NA
NFR-06	Usability	Intuitive UI, Mobile-responsive	Usability	High	Operations Manager	Proposed	NA	NA

* Type Key: -

- Technical
- Integration
- Security
- Audit
- Performance
- Capacity
- Availability
- Reliability
- Recovery
- Compatibility
- Maintainability
- Usability

** Status Key: -

- Proposed
- Accepted

<Non Functional requirements checklist:-

TYPE	SAMPLE DETAIL
SECURITY	
Login Requirements	Access levels, CRUD levels (Create/Retrieve/Update/Delete)
Password Requirements	Length, special characters, expiry, recycling policies
Inactivity Timeouts	Durations, actions
AUDIT	
Audited Elements	What business elements will be audited
Audited Fields	Which data fields will be audited
Audit File Characteristics	Before image, after image, user & time stamp
PEFORMANCE	
Response Times	Application loading, screen open & refresh times
Processing Times	Functions, calculations, imports, exports
Query & Reporting Times	Initial loads & subsequent loads
CAPACITY	
Throughput	How many transactions per hour does the system need to be able to handle?
Storage	How much data does the system need to be able to store?
Year on year growth requirements	
AVAILABILITY	
Hours of Operation	When is it available? Consider weekends, holidays, maintenance times etc
Locations of Operation	Where should it be available from; what are the connection requirements?
RELIABILITY	
Meantime Between Failures	What is the acceptable threshold for down-time eg. one a year, 4,000 hours etc
Meantime To Recovery	If broken, how much time is available to get the system back up again?
INTEGRITY	
Fault Trapping	How to handle electronic interface failures
Bad Data Trapping	Data imports; flag-and-continue or stop the import policies etc
Data Integrity	Referential integrity in DB tables and interfaces
Image Compression & Decompression Standards	
RECOVERY	
Recovery process	How do recoveries work; what is the process?
Recovery Timescales	How long should a recovery take to perform?
Backup Frequencies	How often is the transaction data, set-up data and system (code) backed up?
Backup Generations	What are the requirements for restoring to previous instance(s)?
COMPATIBILITY	
Compatibility With Shared Application	What other systems does it need to talk to?
Compatibility With 3rd Party Applications	What other systems does it have to live with amicably?
Compatibility On Different Operating Systems	What does it have to be able to run on?

Compatibility On Different Platforms?	What are the hardware platforms it needs to work on?
MAINTAINABILITY	
Conformance to Architecture Standards	What are the standards it needs to conform to or have exclusions from?
Conformance to Design Standards	What design standards must be adhered to or exclusions created?
Conformance to Coding Standards	What design standards must be adhered to or exclusions created?
USABILITY	
Look And Feel Standards	Screen element density, layout and flow, colours, UI, metaphors, keyboards and shortcuts
Internationalization/ Localization Requirements	Languages, spellings, keyboards, paper sizes etc

16 Business Impact Assessment

Lens	Key Impacts
Process	Automates 8hr/day monitoring, predictive vs reactive maintenance
People	Engineers focus on fixes (not checks), 2hr dashboard training needed
Customer	99.9% water supply reliability, no outages
Financial	Rs.5lacs/year repair savings, 75% fewer breakdowns
Governance & Risk	Early warnings eliminate surprise failures

17 Costs

Estimated Costs:

- Sensor Installation: Rs.3 Lacs (one – time)
- Cloud/ML platform: Rs. 10K per month
- Development: Rs.5 Lacs (6 months)
- ROI: 6 months (Rs.5 Lacs/year savings)

18 Appendices

Appendix A:

Name	Role	Contact
Aman Jain	Operations Manager	abc@gmail.com
Amit Shah	Maintenance Lead	-

Appendix B:

Pump Failure: Drop in flow/pressure below 80% threshold
ML prediction: Machine learning forecast of failure within 48hrs

Appendix C:

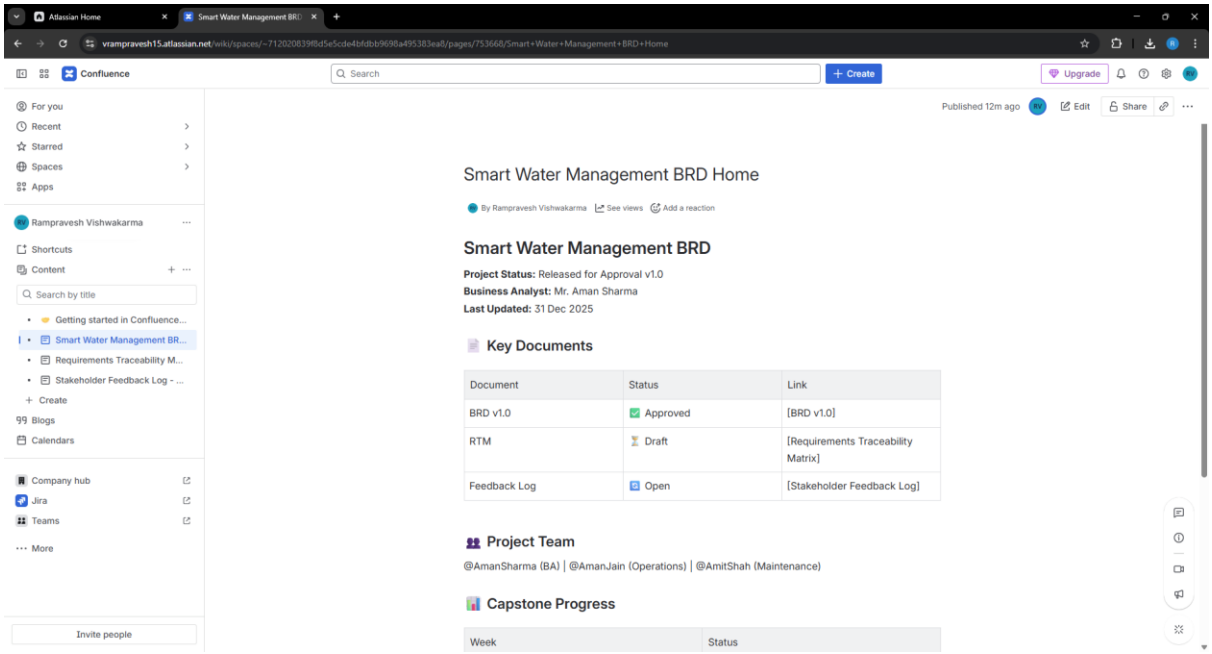
Approval Signatures:

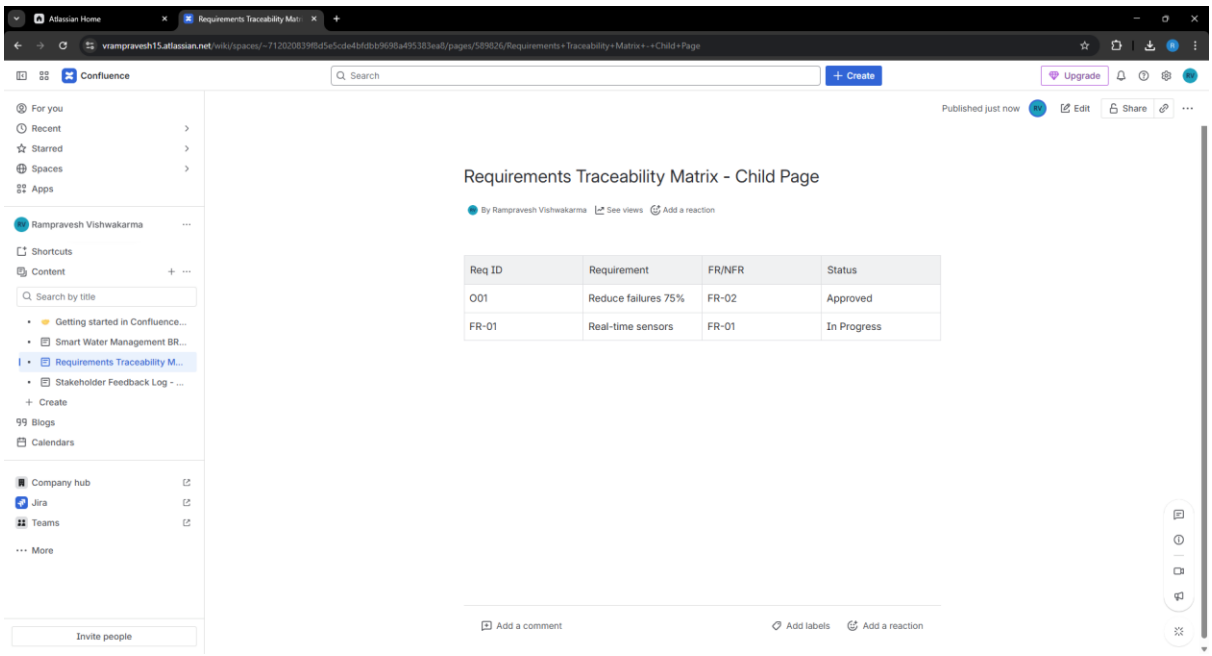
Appendix D:

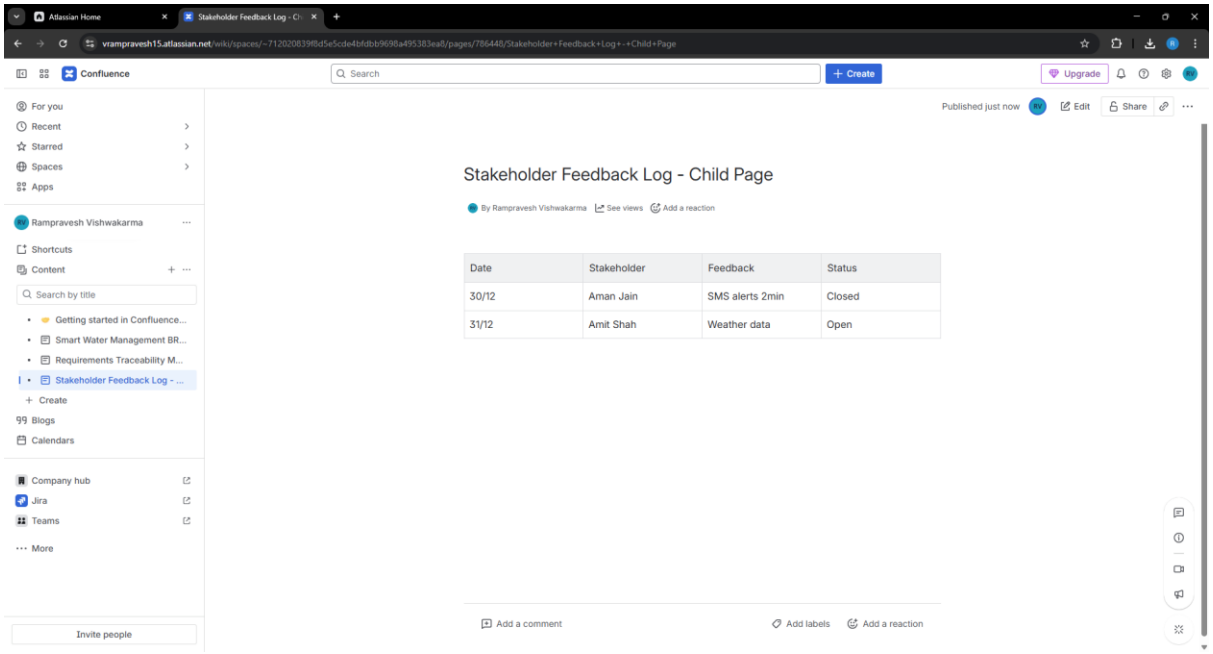
Documentation Repository:
Confluence Space: Smart Water Management BRD
Page link:
https://vrampravesh15.atlassian.net/wiki/spaces/~712020839f8d5e5cde4bfdbb9698a495383ea8/pages/753668/Smart+Water+Management+BRD+Home
Version Control: All Changes Tracked with approvers.

Appendix E:

Review and Handoff the Plan:
Peer Review: Confluence comments
Stakeholder Validation: BRD Approval table
Training: 2hr dashboard session planned
Closure: Lessons Learned in Feedback Log


Smart Water Management BRD Home
Smart Water Management BRD
Project Status: Released for Approval v1.0
Business Analyst: Mr. Aman Sharma
Last Updated: 31 Dec 2025
Key Documents
Document | Status | Link
BRD v1.0 | Approved | [BRD v1.0]
RTM | Draft | [Requirements Traceability Matrix]
Feedback Log | Open | [Stakeholder Feedback Log]
Project Team
@AmanSharma (BA) | @AmanJain (Operations) | @AmitShah (Maintenance)
Capstone Progress
Week | Status


Requirements Traceability Matrix - Child Page
Req ID | Requirement | FR/NFR | Status
001 | Reduce failures 75% | FR-02 | Approved
FR-01 | Real-time sensors | FR-01 | In Progress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Confluence
Search
Create
Upgrade
For you
Recent
Starred
Spaces
Apps
Rampravesh Vishwakarma
Shortcuts
Content
Search by title
Getting started in Confluence...
Smart Water Management BR...
Requirements Traceability M...
Stakeholder Feedback Log - ...
Create
Blogs
Calendars
Company hub
Jira
Teams
More
Invite people
Published just now
Edit
Share
Stakeholder Feedback Log - Child Page
By Rampravesh Vishwakarma
See views
Add a reaction
Date
Stakeholder
Feedback
Status
30/12
Aman Jain
SMS alerts 2min
Closed
31/12
Amit Shah
Weather data
Open
Add a comment
Add labels
Add a reaction

Sign off